Mateusz Janowski

Profile

Computer Science and Intelligent Systems student at AGH University of Krakow with a strong foundation in artificial intelligence and data analysis. Experienced in developing applications based on machine learning and deep neural networks, as well as data engineering projects involving data warehouses, ETL processes, and data analysis tools. Familiar with best practices in optimization and effective teamwork.

Contact Information

mateu.janowski@gmail.com
+48 501 401 921
Krakow, Poland
GitHub
LinkedIn

Personal Information

Date of birth: **21 September 2003**
Nationality: **Polish**
Languages: **Polish** (native), **English** (B2), **German** (A2)

Soft Skills

- teamwork (Agile projects),
- problem-solving,
- project ownership,
- adaptability,
- communication skills,
- time management,
- critical thinking,
- attention to detail,
- willingness to learn

Education

Master's Degree – Computer Science and Intelligent Systems *AGH, Kraków*
2026 – currently
- Specialization: Artificial Intelligence and Data Analysis
- Key topics: AI, KRR, Computational Intelligence, Advanced ML, Advanced Data Mining, Advanced Python, Agent Systems, Cybersecurity

Bachelor's Degree – Computational Engineering *AGH, Kraków*
2022 – 2026
- Key Topics: Modelling, Software Engineering, Exploratory Data Analysis, Knowledge Engineering, Numerical Methods, Deep Neural Networks

Exchange Program (Erasmus+) – AI and Data Science *ISEP, Porto*
2024 – 2025
- Key Topics: Text Mining, Process Mining, ML, Deep Learning, Reinforcement Learning, Advanced Algorithms, Evolutionary Algorithms

Projects

- Agent-Based Predator–Prey Simulation (2026–present) – Multi-agent simulation in Unity modelling predator–prey dynamics. Includes perception, movement, energy consumption, and reproduction. Extended with reinforcement learning (PPO, Unity ML-Agents) to enable adaptive behaviour and emergent strategies.

- Open-Access Research Paper Scraping (2026–present) – Pipeline for extracting structured data from scientific articles (arXiv, PubMed). Includes web scraping, text parsing, and NLP-based section detection (Methods, Risk of Bias).

- Advanced Hospital Management System (2024–2025) – Full-stack platform for medical operation scheduling (.NET, Angular). Role-based system, Prolog-based optimization, and REST API with 3D visualization (Three.js).

- Vegetation Classification with Machine Learning (2024) – ML pipeline for vegetation classification using GAM, Random Forest, and SVM with feature engineering and class balancing.

- BSc Thesis: Dependency Pattern Discovery in Noisy Industrial Data (2025–2026) – Predictive pipeline for industrial time-series using time-lag detection, denoising, and ensemble models (SARIMAX + Gradient Boosting).

Publications and Seminars

- Web Search and Information Retrieval in the Internet (2025) – Seminar presented at ISEP on search engine architectures and models (BM25, DPR, BERT) and their applications in AI systems.

Programming Languages

- Python, R, SQL, C++, Java, C, C#, JavaScript, TypeScript, HTML, CSS, Prolog, Logica, LaTeX

Technologies and Tools

- TensorFlow, Scikit-learn, Jupyter, Azure, AWS, .NET, Angular, Three.js, Git, Visual Studio, CPN Tools

Research Interests

- Machine Learning, Deep Learning, Data Analysis, Data Engineering, Scalable ML Systems, Explainable AI, Data Pipelines and ETL, Applied Artificial Intelligence

Awards

- Scholarship for top-performing students, AGH University of Krakow

Interests

- Martial arts (Kung Fu), football, cycling, climbing, trekking